

SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

Executive Summary

This report compares energy consumption and operational costs between baseline and all applied renovation scenarios from CSV results. Embodied carbon is annualized using an analysis period of 30 years, while operational carbon is reported over 1 year. The construction cost data is from 2026 RSMeans Database. The embodied carbon is calculated using data from EC3 Environmental Product Declaration (EPD) Database. Since RSMeans Database is proprietary, users can adopt customized cost dataset when needed.

Building Information

- Weather File: USA_MN_Duluth.Intl.AP-Duluth.ANGB.727450_TMYx.epw
- Building Name: SmallOffice
- Building Location: Duluth, MN
- Total Floor Area: N/A

Renovation Details by Scenario

Scenario	Renovation Type	Renovation Details
Scenario 1	unknown	Not available in CSV
Scenario 2	unknown	Not available in CSV
Scenario 3	unknown	Not available in CSV
Scenario 4	unknown	Not available in CSV
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Scenario 6	unknown	Not available in CSV
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Scenario 205	unknown	Not available in CSV
Scenario 206	unknown	Not available in CSV

Annual Energy Analysis

Scenario	Metric	Baseline	Renovation	Delta	Savings %
Scenario 1	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 2	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 3	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 4	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 5	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 6	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%

Scenario 7	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 8	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 9	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 10	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 11	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 12	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 13	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 14	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 15	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 16	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 17	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 18	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 19	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 20	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 21	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 22	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 23	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 24	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 25	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 26	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 27	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%

Scenario 28	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 29	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 30	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 31	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 32	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 33	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 34	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 35	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 36	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 37	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 38	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 39	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 40	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 41	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 42	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 43	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 44	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 45	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 46	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 47	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 48	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%

Scenario 49	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 50	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 51	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 52	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 53	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 54	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 55	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 56	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 57	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 58	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 59	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 60	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 61	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 62	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 63	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 64	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 65	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 66	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 67	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 68	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 69	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%

Scenario 70	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 71	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 72	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 73	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 74	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 75	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
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Scenario 80	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
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Scenario 91	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
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Scenario 145	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 146	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 147	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 148	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 149	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 150	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 151	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 152	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 153	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%

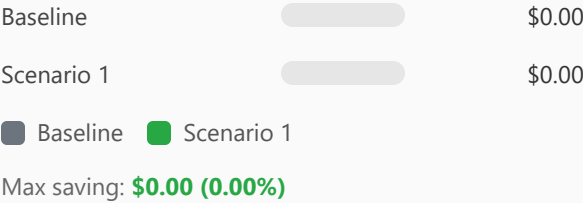
Scenario 154	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 155	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 156	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 157	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 158	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
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Scenario 161	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 162	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 163	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 164	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 165	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 166	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 167	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
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Scenario 175	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
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Scenario 178	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 179	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 180	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 181	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 182	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 183	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 184	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 185	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 186	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 187	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 188	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 189	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 190	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 191	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 192	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 193	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 194	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 195	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%

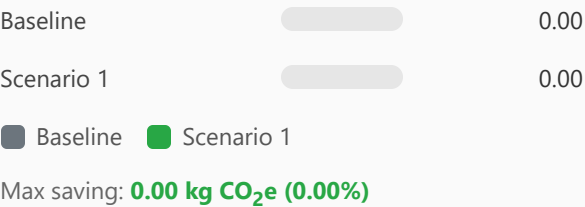
Scenario 196	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 197	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 198	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 199	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 200	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 201	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 202	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 203	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 204	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 205	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%
Scenario 206	Total Site Energy (GJ)	0.00	0.00	0.00	0.00%

Key Performance Metrics

Operational Cost Saving (Scenario - Baseline) (\$/yr) Comparison



Operational Carbon Saving (Scenario - Baseline) (kg CO₂e/yr) Comparison



Min Retrofit Construction Cost

N/A

Construction cost not in CSV

Min Retrofit Embodied Carbon

0.00 kg CO₂e

Embodied carbon intensity (ECI): N/A
Scenario 1

Lowest Retrofit Cost Payback Period

N/A

No valid cost payback scenario

Lowest Retrofit Carbon Payback Period

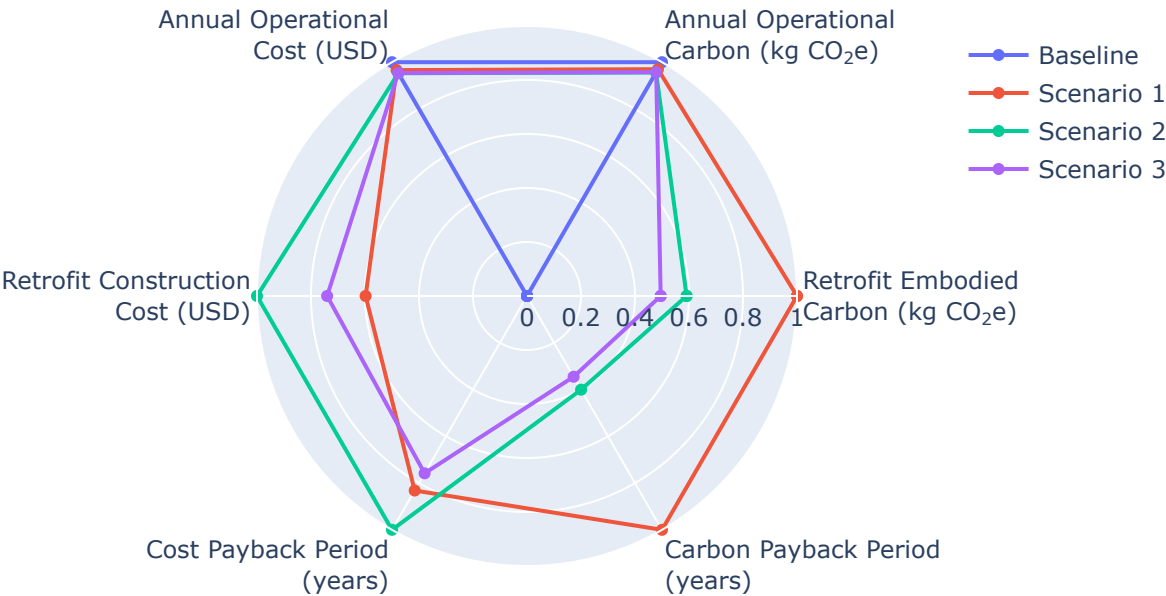
N/A

No valid carbon payback scenario

Comparative Visualizations between Renovation Scenarios

Spider Chart Visualization

Parametric Scenario Comparison (Normalized Spider Chart)



Payback Period of the Retrofit

Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) (\$/yr)).

Scenario 1		N/A
Scenario 2		N/A
Scenario 3		N/A

Carbon Payback Period

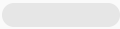
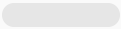
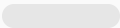
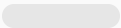
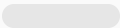
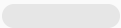
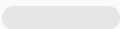
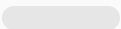
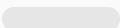
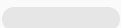
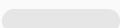
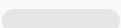
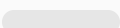
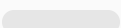
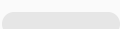
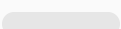
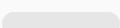
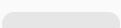
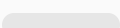
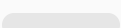
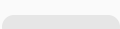
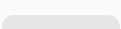
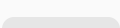
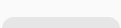
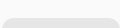
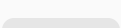
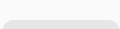
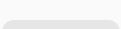
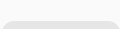
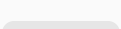
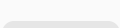
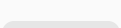
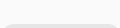
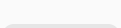
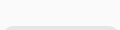
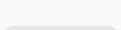
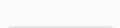
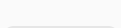
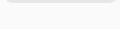
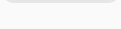
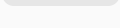
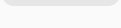
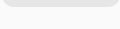
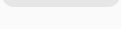
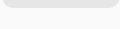
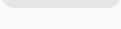
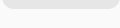
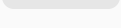
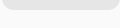
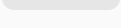
Payback = retrofit embodied carbon / abs(Operational Carbon Saving (Scenario - Baseline) (kg CO₂e/yr)).

Scenario 1		N/A
Scenario 2		N/A
Scenario 3		N/A

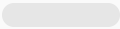
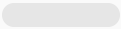
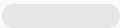
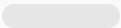
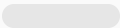
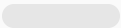
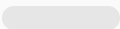
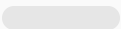
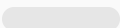
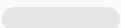
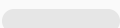
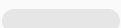
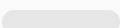
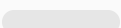
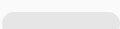
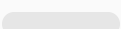
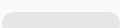
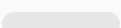
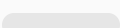
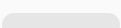
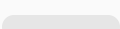
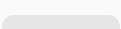
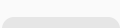
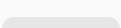
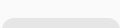
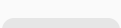
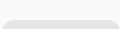
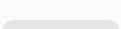
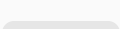
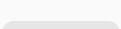
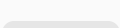
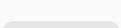
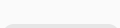
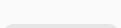
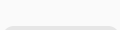
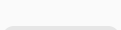
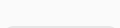
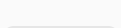
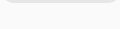
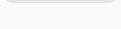
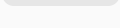
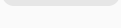
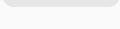
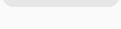
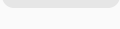
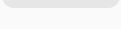
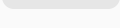
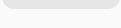
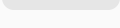
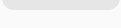
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Scenario 7		N/A	Scenario 7		N/A
Scenario 8		N/A	Scenario 8		N/A
Scenario 9		N/A	Scenario 9		N/A
Scenario 10		N/A	Scenario 10		N/A
Scenario 11		N/A	Scenario 11		N/A
Scenario 12		N/A	Scenario 12		N/A
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Scenario 15		N/A	Scenario 15		N/A
Scenario 16		N/A	Scenario 16		N/A
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Scenario 18		N/A	Scenario 18		N/A
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Scenario 26		N/A	Scenario 26		N/A
Scenario 27		N/A	Scenario 27		N/A
Scenario 28		N/A	Scenario 28		N/A
Scenario 29		N/A	Scenario 29		N/A
Scenario 30		N/A	Scenario 30		N/A
Scenario 31		N/A	Scenario 31		N/A
Scenario 32		N/A	Scenario 32		N/A
Scenario 33		N/A	Scenario 33		N/A
Scenario 34		N/A	Scenario 34		N/A
Scenario 35		N/A	Scenario 35		N/A
Scenario 36		N/A	Scenario 36		N/A

Scenario 37		N/A	Scenario 37		N/A
Scenario 38		N/A	Scenario 38		N/A
Scenario 39		N/A	Scenario 39		N/A
Scenario 40		N/A	Scenario 40		N/A
Scenario 41		N/A	Scenario 41		N/A
Scenario 42		N/A	Scenario 42		N/A
Scenario 43		N/A	Scenario 43		N/A
Scenario 44		N/A	Scenario 44		N/A
Scenario 45		N/A	Scenario 45		N/A
Scenario 46		N/A	Scenario 46		N/A
Scenario 47		N/A	Scenario 47		N/A
Scenario 48		N/A	Scenario 48		N/A
Scenario 49		N/A	Scenario 49		N/A
Scenario 50		N/A	Scenario 50		N/A
Scenario 51		N/A	Scenario 51		N/A
Scenario 52		N/A	Scenario 52		N/A
Scenario 53		N/A	Scenario 53		N/A
Scenario 54		N/A	Scenario 54		N/A
Scenario 55		N/A	Scenario 55		N/A
Scenario 56		N/A	Scenario 56		N/A
Scenario 57		N/A	Scenario 57		N/A
Scenario 58		N/A	Scenario 58		N/A
Scenario 59		N/A	Scenario 59		N/A
Scenario 60		N/A	Scenario 60		N/A
Scenario 61		N/A	Scenario 61		N/A
Scenario 62		N/A	Scenario 62		N/A
Scenario 63		N/A	Scenario 63		N/A
Scenario 64		N/A	Scenario 64		N/A
Scenario 65		N/A	Scenario 65		N/A
Scenario 66		N/A	Scenario 66		N/A
Scenario 67		N/A	Scenario 67		N/A
Scenario 68		N/A	Scenario 68		N/A
Scenario 69		N/A	Scenario 69		N/A

Scenario 70		N/A	Scenario 70		N/A
Scenario 71		N/A	Scenario 71		N/A
Scenario 72		N/A	Scenario 72		N/A
Scenario 73		N/A	Scenario 73		N/A
Scenario 74		N/A	Scenario 74		N/A
Scenario 75		N/A	Scenario 75		N/A
Scenario 76		N/A	Scenario 76		N/A
Scenario 77		N/A	Scenario 77		N/A
Scenario 78		N/A	Scenario 78		N/A
Scenario 79		N/A	Scenario 79		N/A
Scenario 80		N/A	Scenario 80		N/A
Scenario 81		N/A	Scenario 81		N/A
Scenario 82		N/A	Scenario 82		N/A
Scenario 83		N/A	Scenario 83		N/A
Scenario 84		N/A	Scenario 84		N/A
Scenario 85		N/A	Scenario 85		N/A
Scenario 86		N/A	Scenario 86		N/A
Scenario 87		N/A	Scenario 87		N/A
Scenario 88		N/A	Scenario 88		N/A
Scenario 89		N/A	Scenario 89		N/A
Scenario 90		N/A	Scenario 90		N/A
Scenario 91		N/A	Scenario 91		N/A
Scenario 92		N/A	Scenario 92		N/A
Scenario 93		N/A	Scenario 93		N/A
Scenario 94		N/A	Scenario 94		N/A
Scenario 95		N/A	Scenario 95		N/A
Scenario 96		N/A	Scenario 96		N/A
Scenario 97		N/A	Scenario 97		N/A
Scenario 98		N/A	Scenario 98		N/A
Scenario 99		N/A	Scenario 99		N/A
Scenario 100		N/A	Scenario 100		N/A
Scenario 101		N/A	Scenario 101		N/A
Scenario 102		N/A	Scenario 102		N/A

Scenario 103		N/A	Scenario 103		N/A
Scenario 104		N/A	Scenario 104		N/A
Scenario 105		N/A	Scenario 105		N/A
Scenario 106		N/A	Scenario 106		N/A
Scenario 107		N/A	Scenario 107		N/A
Scenario 108		N/A	Scenario 108		N/A
Scenario 109		N/A	Scenario 109		N/A
Scenario 110		N/A	Scenario 110		N/A
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Scenario 112		N/A	Scenario 112		N/A
Scenario 113		N/A	Scenario 113		N/A
Scenario 114		N/A	Scenario 114		N/A
Scenario 115		N/A	Scenario 115		N/A
Scenario 116		N/A	Scenario 116		N/A
Scenario 117		N/A	Scenario 117		N/A
Scenario 118		N/A	Scenario 118		N/A
Scenario 119		N/A	Scenario 119		N/A
Scenario 120		N/A	Scenario 120		N/A
Scenario 121		N/A	Scenario 121		N/A
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Scenario 123		N/A	Scenario 123		N/A
Scenario 124		N/A	Scenario 124		N/A
Scenario 125		N/A	Scenario 125		N/A
Scenario 126		N/A	Scenario 126		N/A
Scenario 127		N/A	Scenario 127		N/A
Scenario 128		N/A	Scenario 128		N/A
Scenario 129		N/A	Scenario 129		N/A
Scenario 130		N/A	Scenario 130		N/A
Scenario 131		N/A	Scenario 131		N/A
Scenario 132		N/A	Scenario 132		N/A
Scenario 133		N/A	Scenario 133		N/A
Scenario 134		N/A	Scenario 134		N/A
Scenario 135		N/A	Scenario 135		N/A

Scenario 136		N/A	Scenario 136		N/A
Scenario 137		N/A	Scenario 137		N/A
Scenario 138		N/A	Scenario 138		N/A
Scenario 139		N/A	Scenario 139		N/A
Scenario 140		N/A	Scenario 140		N/A
Scenario 141		N/A	Scenario 141		N/A
Scenario 142		N/A	Scenario 142		N/A
Scenario 143		N/A	Scenario 143		N/A
Scenario 144		N/A	Scenario 144		N/A
Scenario 145		N/A	Scenario 145		N/A
Scenario 146		N/A	Scenario 146		N/A
Scenario 147		N/A	Scenario 147		N/A
Scenario 148		N/A	Scenario 148		N/A
Scenario 149		N/A	Scenario 149		N/A
Scenario 150		N/A	Scenario 150		N/A
Scenario 151		N/A	Scenario 151		N/A
Scenario 152		N/A	Scenario 152		N/A
Scenario 153		N/A	Scenario 153		N/A
Scenario 154		N/A	Scenario 154		N/A
Scenario 155		N/A	Scenario 155		N/A
Scenario 156		N/A	Scenario 156		N/A
Scenario 157		N/A	Scenario 157		N/A
Scenario 158		N/A	Scenario 158		N/A
Scenario 159		N/A	Scenario 159		N/A
Scenario 160		N/A	Scenario 160		N/A
Scenario 161		N/A	Scenario 161		N/A
Scenario 162		N/A	Scenario 162		N/A
Scenario 163		N/A	Scenario 163		N/A
Scenario 164		N/A	Scenario 164		N/A
Scenario 165		N/A	Scenario 165		N/A
Scenario 166		N/A	Scenario 166		N/A
Scenario 167		N/A	Scenario 167		N/A
Scenario 168		N/A	Scenario 168		N/A

Scenario 169		N/A	Scenario 169		N/A
Scenario 170		N/A	Scenario 170		N/A
Scenario 171		N/A	Scenario 171		N/A
Scenario 172		N/A	Scenario 172		N/A
Scenario 173		N/A	Scenario 173		N/A
Scenario 174		N/A	Scenario 174		N/A
Scenario 175		N/A	Scenario 175		N/A
Scenario 176		N/A	Scenario 176		N/A
Scenario 177		N/A	Scenario 177		N/A
Scenario 178		N/A	Scenario 178		N/A
Scenario 179		N/A	Scenario 179		N/A
Scenario 180		N/A	Scenario 180		N/A
Scenario 181		N/A	Scenario 181		N/A
Scenario 182		N/A	Scenario 182		N/A
Scenario 183		N/A	Scenario 183		N/A
Scenario 184		N/A	Scenario 184		N/A
Scenario 185		N/A	Scenario 185		N/A
Scenario 186		N/A	Scenario 186		N/A
Scenario 187		N/A	Scenario 187		N/A
Scenario 188		N/A	Scenario 188		N/A
Scenario 189		N/A	Scenario 189		N/A
Scenario 190		N/A	Scenario 190		N/A
Scenario 191		N/A	Scenario 191		N/A
Scenario 192		N/A	Scenario 192		N/A
Scenario 193		N/A	Scenario 193		N/A
Scenario 194		N/A	Scenario 194		N/A
Scenario 195		N/A	Scenario 195		N/A
Scenario 196		N/A	Scenario 196		N/A
Scenario 197		N/A	Scenario 197		N/A
Scenario 198		N/A	Scenario 198		N/A
Scenario 199		N/A	Scenario 199		N/A
Scenario 200		N/A	Scenario 200		N/A
Scenario 201		N/A	Scenario 201		N/A

Scenario 202		N/A	Scenario 202		N/A
Scenario 203		N/A	Scenario 203		N/A
Scenario 204		N/A	Scenario 204		N/A
Scenario 205		N/A	Scenario 205		N/A
Scenario 206		N/A	Scenario 206		N/A

Material List

No material consumption data found for renovation scenarios.

Result Summary

Per-scenario retrofit contribution breakdown. Each scenario includes two pie charts: cost and embodied carbon by retrofit measure.

Scenario 1

Cost breakdown by retrofit measure

No retrofit measure contribution data.

Carbon breakdown by retrofit measure

No retrofit measure contribution data.

Scenario 2

Cost breakdown by retrofit measure

No retrofit measure contribution data.

Carbon breakdown by retrofit measure

No retrofit measure contribution data.

Scenario 3

Cost breakdown by retrofit measure

No retrofit measure contribution data.

Carbon breakdown by retrofit measure

No retrofit measure contribution data.

Scenario 4

Cost breakdown by retrofit measure

Carbon breakdown by retrofit measure

No retrofit measure contribution data.

No retrofit measure contribution data.

Scenario 5

Cost breakdown by retrofit measure
No retrofit measure contribution data.

Carbon breakdown by retrofit measure
No retrofit measure contribution data.

Scenario 6

Cost breakdown by retrofit measure
No retrofit measure contribution data.

Carbon breakdown by retrofit measure
No retrofit measure contribution data.

Scenario 7

Cost breakdown by retrofit measure
No retrofit measure contribution data.

Carbon breakdown by retrofit measure
No retrofit measure contribution data.

Scenario 8

Cost breakdown by retrofit measure
No retrofit measure contribution data.

Carbon breakdown by retrofit measure
No retrofit measure contribution data.

Scenario 9

Cost breakdown by retrofit measure
No retrofit measure contribution data.

Carbon breakdown by retrofit measure
No retrofit measure contribution data.

Scenario 10

Cost breakdown by retrofit measure
No retrofit measure contribution data.

Carbon breakdown by retrofit measure
No retrofit measure contribution data.

Scenario 11

Cost breakdown by retrofit measure*No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 12****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 13****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 14****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 15****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 16****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 17****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.*

Scenario 18**Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 19****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 20****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 21****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 22****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 23****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 24**

Cost breakdown by retrofit measure*No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 25****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 26****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 27****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 28****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 29****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 30****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.*

Scenario 31**Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 32****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 33****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 34****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 35****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 36****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 37**

Cost breakdown by retrofit measure*No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 38****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 39****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 40****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 41****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 42****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 43****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.*

Scenario 44**Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 45****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 46****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 47****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 48****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 49****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 50**

Cost breakdown by retrofit measure*No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 51****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 52****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 53****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 54****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 55****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 56****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.*

Scenario 58**Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 59****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 60****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 61****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 62****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 63****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.*

Scenario 64**Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 65****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 66****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 67****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 68****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 69****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 70**

Cost breakdown by retrofit measure*No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 71****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 72****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 73****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 74****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 75****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 76****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.*

Scenario 77**Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 78****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 79****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 80****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 81****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 82****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 83**

Cost breakdown by retrofit measure*No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 84****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 85****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 86****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 87****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 88****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 89****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.*

Scenario 91**Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 92****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 93****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 94****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 95****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 96**

Cost breakdown by retrofit measure*No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 97****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 98****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 99****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 100****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 101****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 102****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.*

Scenario 103

Cost breakdown by retrofit measure

No retrofit measure contribution data.

Carbon breakdown by retrofit measure

No retrofit measure contribution data.

Scenario 104

Cost breakdown by retrofit measure

No retrofit measure contribution data.

Carbon breakdown by retrofit measure

No retrofit measure contribution data.

Scenario 105

Cost breakdown by retrofit measure

No retrofit measure contribution data.

Carbon breakdown by retrofit measure

No retrofit measure contribution data.

Scenario 106

Cost breakdown by retrofit measure

No retrofit measure contribution data.

Carbon breakdown by retrofit measure

No retrofit measure contribution data.

Scenario 107

Cost breakdown by retrofit measure

No retrofit measure contribution data.

Carbon breakdown by retrofit measure

No retrofit measure contribution data.

Scenario 108

Cost breakdown by retrofit measure

No retrofit measure contribution data.

Carbon breakdown by retrofit measure

No retrofit measure contribution data.

Scenario 109

Cost breakdown by retrofit measure*No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 110****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 111****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 112****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 113****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 114****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 115****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.*

Scenario 116**Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 117****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 118****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 119****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 120****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 121****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 122**

Cost breakdown by retrofit measure*No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 123****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 124****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 125****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 126****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 127****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 128****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.*

Scenario 129**Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 130****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 131****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 132****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 133****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 134****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 135**

Cost breakdown by retrofit measure*No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 136****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 137****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 138****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 139****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 140****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 141****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.*

Scenario 142**Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 143****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 144****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 145****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 146****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 147****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 148**

Cost breakdown by retrofit measure*No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 149****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 150****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 151****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 152****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 153****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 154****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.*

Scenario 155**Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 156****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 157****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 158****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 159****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 160****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 161**

Cost breakdown by retrofit measure*No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 162****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 163****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 164****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 165****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 166****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 167****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.*

Scenario 168**Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 169****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 170****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 171****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 172****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 173****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 174**

Cost breakdown by retrofit measure*No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 175****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 176****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 177****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 178****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 179****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 180****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.*

Scenario 182**Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 183****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 184****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 185****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 186****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 187**

Cost breakdown by retrofit measure*No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 188****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 189****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 190****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 191****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 192****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 193****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.*

Scenario 194**Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 195****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 196****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 197****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 198****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 199****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 200**

Cost breakdown by retrofit measure*No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 201****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 202****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 203****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 204****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 205****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.***Scenario 206****Cost breakdown by retrofit measure***No retrofit measure contribution data.***Carbon breakdown by retrofit measure***No retrofit measure contribution data.*

Scenario	Retrofit Embodied Carbon (kg CO ₂ e)	Retrofit Construction Cost (USD)	Annual Operational Carbon (kg CO ₂ e)	Annual Operational Cost (USD)
Scenario 1	0.00	N/A	0.00	\$0.00
Scenario 2	0.00	N/A	0.00	\$0.00
Scenario 3	0.00	N/A	0.00	\$0.00
Scenario 4	0.00	N/A	0.00	\$0.00
Scenario 5	0.00	N/A	0.00	\$0.00
Scenario 6	0.00	N/A	0.00	\$0.00
Scenario 7	0.00	N/A	0.00	\$0.00
Scenario 8	0.00	N/A	0.00	\$0.00
Scenario 9	0.00	N/A	0.00	\$0.00
Scenario 10	0.00	N/A	0.00	\$0.00
Scenario 11	0.00	N/A	0.00	\$0.00
Scenario 12	0.00	N/A	0.00	\$0.00
Scenario 13	0.00	N/A	0.00	\$0.00
Scenario 14	0.00	N/A	0.00	\$0.00
Scenario 15	0.00	N/A	0.00	\$0.00
Scenario 16	0.00	N/A	0.00	\$0.00
Scenario 17	0.00	N/A	0.00	\$0.00
Scenario 18	0.00	N/A	0.00	\$0.00

Scenario 19	0.00	N/A	0.00	\$0.00
Scenario 20	0.00	N/A	0.00	\$0.00
Scenario 21	0.00	N/A	0.00	\$0.00
Scenario 22	0.00	N/A	0.00	\$0.00
Scenario 23	0.00	N/A	0.00	\$0.00
Scenario 24	0.00	N/A	0.00	\$0.00
Scenario 25	0.00	N/A	0.00	\$0.00
Scenario 26	0.00	N/A	0.00	\$0.00
Scenario 27	0.00	N/A	0.00	\$0.00
Scenario 28	0.00	N/A	0.00	\$0.00
Scenario 29	0.00	N/A	0.00	\$0.00
Scenario 30	0.00	N/A	0.00	\$0.00
Scenario 31	0.00	N/A	0.00	\$0.00
Scenario 32	0.00	N/A	0.00	\$0.00
Scenario 33	0.00	N/A	0.00	\$0.00
Scenario 34	0.00	N/A	0.00	\$0.00
Scenario 35	0.00	N/A	0.00	\$0.00
Scenario 36	0.00	N/A	0.00	\$0.00
Scenario 37	0.00	N/A	0.00	\$0.00
Scenario 38	0.00	N/A	0.00	\$0.00
Scenario 39	0.00	N/A	0.00	\$0.00

Scenario 40	0.00	N/A	0.00	\$0.00
Scenario 41	0.00	N/A	0.00	\$0.00
Scenario 42	0.00	N/A	0.00	\$0.00
Scenario 43	0.00	N/A	0.00	\$0.00
Scenario 44	0.00	N/A	0.00	\$0.00
Scenario 45	0.00	N/A	0.00	\$0.00
Scenario 46	0.00	N/A	0.00	\$0.00
Scenario 47	0.00	N/A	0.00	\$0.00
Scenario 48	0.00	N/A	0.00	\$0.00
Scenario 49	0.00	N/A	0.00	\$0.00
Scenario 50	0.00	N/A	0.00	\$0.00
Scenario 51	0.00	N/A	0.00	\$0.00
Scenario 52	0.00	N/A	0.00	\$0.00
Scenario 53	0.00	N/A	0.00	\$0.00
Scenario 54	0.00	N/A	0.00	\$0.00
Scenario 55	0.00	N/A	0.00	\$0.00
Scenario 56	0.00	N/A	0.00	\$0.00
Scenario 57	0.00	N/A	0.00	\$0.00
Scenario 58	0.00	N/A	0.00	\$0.00
Scenario 59	0.00	N/A	0.00	\$0.00
Scenario 60	0.00	N/A	0.00	\$0.00

Scenario 61	0.00	N/A	0.00	\$0.00
Scenario 62	0.00	N/A	0.00	\$0.00
Scenario 63	0.00	N/A	0.00	\$0.00
Scenario 64	0.00	N/A	0.00	\$0.00
Scenario 65	0.00	N/A	0.00	\$0.00
Scenario 66	0.00	N/A	0.00	\$0.00
Scenario 67	0.00	N/A	0.00	\$0.00
Scenario 68	0.00	N/A	0.00	\$0.00
Scenario 69	0.00	N/A	0.00	\$0.00
Scenario 70	0.00	N/A	0.00	\$0.00
Scenario 71	0.00	N/A	0.00	\$0.00
Scenario 72	0.00	N/A	0.00	\$0.00
Scenario 73	0.00	N/A	0.00	\$0.00
Scenario 74	0.00	N/A	0.00	\$0.00
Scenario 75	0.00	N/A	0.00	\$0.00
Scenario 76	0.00	N/A	0.00	\$0.00
Scenario 77	0.00	N/A	0.00	\$0.00
Scenario 78	0.00	N/A	0.00	\$0.00
Scenario 79	0.00	N/A	0.00	\$0.00
Scenario 80	0.00	N/A	0.00	\$0.00
Scenario 81	0.00	N/A	0.00	\$0.00

Scenario 82	0.00	N/A	0.00	\$0.00
Scenario 83	0.00	N/A	0.00	\$0.00
Scenario 84	0.00	N/A	0.00	\$0.00
Scenario 85	0.00	N/A	0.00	\$0.00
Scenario 86	0.00	N/A	0.00	\$0.00
Scenario 87	0.00	N/A	0.00	\$0.00
Scenario 88	0.00	N/A	0.00	\$0.00
Scenario 89	0.00	N/A	0.00	\$0.00
Scenario 90	0.00	N/A	0.00	\$0.00
Scenario 91	0.00	N/A	0.00	\$0.00
Scenario 92	0.00	N/A	0.00	\$0.00
Scenario 93	0.00	N/A	0.00	\$0.00
Scenario 94	0.00	N/A	0.00	\$0.00
Scenario 95	0.00	N/A	0.00	\$0.00
Scenario 96	0.00	N/A	0.00	\$0.00
Scenario 97	0.00	N/A	0.00	\$0.00
Scenario 98	0.00	N/A	0.00	\$0.00
Scenario 99	0.00	N/A	0.00	\$0.00
Scenario 100	0.00	N/A	0.00	\$0.00
Scenario 101	0.00	N/A	0.00	\$0.00
Scenario 102	0.00	N/A	0.00	\$0.00

Scenario 103	0.00	N/A	0.00	\$0.00
Scenario 104	0.00	N/A	0.00	\$0.00
Scenario 105	0.00	N/A	0.00	\$0.00
Scenario 106	0.00	N/A	0.00	\$0.00
Scenario 107	0.00	N/A	0.00	\$0.00
Scenario 108	0.00	N/A	0.00	\$0.00
Scenario 109	0.00	N/A	0.00	\$0.00
Scenario 110	0.00	N/A	0.00	\$0.00
Scenario 111	0.00	N/A	0.00	\$0.00
Scenario 112	0.00	N/A	0.00	\$0.00
Scenario 113	0.00	N/A	0.00	\$0.00
Scenario 114	0.00	N/A	0.00	\$0.00
Scenario 115	0.00	N/A	0.00	\$0.00
Scenario 116	0.00	N/A	0.00	\$0.00
Scenario 117	0.00	N/A	0.00	\$0.00
Scenario 118	0.00	N/A	0.00	\$0.00
Scenario 119	0.00	N/A	0.00	\$0.00
Scenario 120	0.00	N/A	0.00	\$0.00
Scenario 121	0.00	N/A	0.00	\$0.00
Scenario 122	0.00	N/A	0.00	\$0.00
Scenario 123	0.00	N/A	0.00	\$0.00

Scenario 124	0.00	N/A	0.00	\$0.00
Scenario 125	0.00	N/A	0.00	\$0.00
Scenario 126	0.00	N/A	0.00	\$0.00
Scenario 127	0.00	N/A	0.00	\$0.00
Scenario 128	0.00	N/A	0.00	\$0.00
Scenario 129	0.00	N/A	0.00	\$0.00
Scenario 130	0.00	N/A	0.00	\$0.00
Scenario 131	0.00	N/A	0.00	\$0.00
Scenario 132	0.00	N/A	0.00	\$0.00
Scenario 133	0.00	N/A	0.00	\$0.00
Scenario 134	0.00	N/A	0.00	\$0.00
Scenario 135	0.00	N/A	0.00	\$0.00
Scenario 136	0.00	N/A	0.00	\$0.00
Scenario 137	0.00	N/A	0.00	\$0.00
Scenario 138	0.00	N/A	0.00	\$0.00
Scenario 139	0.00	N/A	0.00	\$0.00
Scenario 140	0.00	N/A	0.00	\$0.00
Scenario 141	0.00	N/A	0.00	\$0.00
Scenario 142	0.00	N/A	0.00	\$0.00
Scenario 143	0.00	N/A	0.00	\$0.00
Scenario 144	0.00	N/A	0.00	\$0.00

Scenario 145	0.00	N/A	0.00	\$0.00
Scenario 146	0.00	N/A	0.00	\$0.00
Scenario 147	0.00	N/A	0.00	\$0.00
Scenario 148	0.00	N/A	0.00	\$0.00
Scenario 149	0.00	N/A	0.00	\$0.00
Scenario 150	0.00	N/A	0.00	\$0.00
Scenario 151	0.00	N/A	0.00	\$0.00
Scenario 152	0.00	N/A	0.00	\$0.00
Scenario 153	0.00	N/A	0.00	\$0.00
Scenario 154	0.00	N/A	0.00	\$0.00
Scenario 155	0.00	N/A	0.00	\$0.00
Scenario 156	0.00	N/A	0.00	\$0.00
Scenario 157	0.00	N/A	0.00	\$0.00
Scenario 158	0.00	N/A	0.00	\$0.00
Scenario 159	0.00	N/A	0.00	\$0.00
Scenario 160	0.00	N/A	0.00	\$0.00
Scenario 161	0.00	N/A	0.00	\$0.00
Scenario 162	0.00	N/A	0.00	\$0.00
Scenario 163	0.00	N/A	0.00	\$0.00
Scenario 164	0.00	N/A	0.00	\$0.00
Scenario 165	0.00	N/A	0.00	\$0.00

Scenario 166	0.00	N/A	0.00	\$0.00
Scenario 167	0.00	N/A	0.00	\$0.00
Scenario 168	0.00	N/A	0.00	\$0.00
Scenario 169	0.00	N/A	0.00	\$0.00
Scenario 170	0.00	N/A	0.00	\$0.00
Scenario 171	0.00	N/A	0.00	\$0.00
Scenario 172	0.00	N/A	0.00	\$0.00
Scenario 173	0.00	N/A	0.00	\$0.00
Scenario 174	0.00	N/A	0.00	\$0.00
Scenario 175	0.00	N/A	0.00	\$0.00
Scenario 176	0.00	N/A	0.00	\$0.00
Scenario 177	0.00	N/A	0.00	\$0.00
Scenario 178	0.00	N/A	0.00	\$0.00
Scenario 179	0.00	N/A	0.00	\$0.00
Scenario 180	0.00	N/A	0.00	\$0.00
Scenario 181	0.00	N/A	0.00	\$0.00
Scenario 182	0.00	N/A	0.00	\$0.00
Scenario 183	0.00	N/A	0.00	\$0.00
Scenario 184	0.00	N/A	0.00	\$0.00
Scenario 185	0.00	N/A	0.00	\$0.00
Scenario 186	0.00	N/A	0.00	\$0.00

Scenario 187	0.00	N/A	0.00	\$0.00
Scenario 188	0.00	N/A	0.00	\$0.00
Scenario 189	0.00	N/A	0.00	\$0.00
Scenario 190	0.00	N/A	0.00	\$0.00
Scenario 191	0.00	N/A	0.00	\$0.00
Scenario 192	0.00	N/A	0.00	\$0.00
Scenario 193	0.00	N/A	0.00	\$0.00
Scenario 194	0.00	N/A	0.00	\$0.00
Scenario 195	0.00	N/A	0.00	\$0.00
Scenario 196	0.00	N/A	0.00	\$0.00
Scenario 197	0.00	N/A	0.00	\$0.00
Scenario 198	0.00	N/A	0.00	\$0.00
Scenario 199	0.00	N/A	0.00	\$0.00
Scenario 200	0.00	N/A	0.00	\$0.00
Scenario 201	0.00	N/A	0.00	\$0.00
Scenario 202	0.00	N/A	0.00	\$0.00
Scenario 203	0.00	N/A	0.00	\$0.00
Scenario 204	0.00	N/A	0.00	\$0.00
Scenario 205	0.00	N/A	0.00	\$0.00
Scenario 206	0.00	N/A	0.00	\$0.00

Generated: May 01, 2026 | Report Type: Retrofit Impact Analysis | Run: run_test_008

